HIVEBITS · DRAFT

Pollinator Nodes

Small houses for wild bees with sensors inside. Each one measures local biodiversity where it stands. Together they make a map. Anyone, anywhere, can put one up, and it joins the HiveBits network.

FOR	INSIDE	OUTSIDE	NETWORK
Mason bees (Osmia), wild pollinators	Temp, humidity, entrance counter, daily photo	Temp, humidity, pressure, light, rainfall	On-chain ID per house, rewards for uptime

Where this comes from

Many people asked us for indoor hives, so they could have bees at home and be part of HiveBits. We do not want to do that. Bees belong outside. But the question was right: people want a way to join that is not only buying honey or a token.

This is that way. A house for wild bees you put in your garden, on your balcony, or at your company. It does not need a beekeeper. It needs sun, a wall, and WiFi.

Why wild bees

Honey bees are one species. There are hundreds of wild bee species around every apiary and they do a big part of the pollination. Almost nobody measures them. Bee hotels are sold everywhere, but a study of 600 of them found more wasps than bees inside, and no data that they help at all.

Research also says honey bees and wild bees share the same flowers, and that too many hives in one place can push wild bees out. A beekeeper who wants to do this right needs to know what is around the apiary. We have 300 hives on our farm. We want that number to be in balance with everything that flies around it, and we want to show it, not just say it.

MacIvor and Packer, PLoS ONE 2015. Henry and Rodet, Scientific Reports 2018. Pasquali et al., Current Biology 2025.

How it fits HiveBits

HiveBits already tracks our hives: humidity, temperature, weight, with our own sensors. The pollinator house is the same system in a smaller box, pointed at the bees we do not keep. First houses go next to our hives on the farm, so every HiveBits farm gets a biodiversity reading around it. Then houses go to people and companies, and the map grows beyond our farms.

For HiveBits this means: more devices in the network, a farm that can show it is pollinator positive with data, and a product for people who cannot keep bees.

What a host gets

- The house with sensors and the first bees (cocoons), so year one is not empty.
- The house in the app: live data, daily photo, "first bee moved in" messages, and your spot on the biodiversity map.
- Network rewards for keeping the house online and sending data.
- A yearly service: send the nesting block back, get it cleaned and refilled.
- Companies also get a biodiversity report for ESG and a live page for employees.

Who pays

- **Companies** that adopt houses for ESG. This works today: 3Bee in Italy sells it to 500+ brands.
- **Food brands and growers** that need proof their supply chain is pollinator safe. M&S and Importaco already pay for pollinator sensors on farms.
- **Later:** certification and EU nature credits, which will need proof that is not collected by hand.

Raw weather data alone has no buyer, WeatherXM showed that. Pollinator activity with weather next to it, for a specific place, does. And the bigger the map, the more it is worth.

Why us

We already have the farm, the hives, our own sensors, the team, and 150 people who funded us through MetaDAO. This is the next step of the same network, not a new company.

What we ask

- Tell us if you would host one, at home or at your company. Share this with people who would.
- Feedback on the design before we build the first batch.
- We plan to raise for the first batch. Details to follow.

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